



Code Mix TTS: An Approach to Infer Human Like Speech for Multi-Lingual Input Texts

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Abstract

We have come a far way in terms of producing high quality, human like audios for given input texts using TTS or Text to Speech Systems. We simply provide the TTS system with an input text and it generates an audio output. But, if we look at it closely, we provide the input text in a particular language only. However, as humans, while interacting with each other, do we actually speak in just one particular language? The answer is No, in most cases. Our speech in general, specially in multiple language countries, is a mix of native language of the specific country and a popular language (mostly English) and sometimes the other way round where we mix a bit of the native language words in our English conversation. In this paper, we talk about an approach to achieve the similar kind of "Code Mix" inference output for "Multi-Lingual" input texts from our TTS system without the need of any additional training data or fine tuning.

Index Terms: TTS, Multi-Lingual Text, StyleTTS2, Automated-MOS

1. Introduction

In this modern day and age, TTS is one of the most sought after pieces of technology that uses AI, more precisely Neural Networks to produce audio for a given Text Input. If you have ever used a modern TTS system, you would know that you have to typically select the language, the speaker, the audio type, the model you want to use for inference and last and most importantly provide your input text in the language selected, and voila, an audio is generated for you. The audio though, human like might sound robotic to some listener because, we humans, in general do not speak in a single language. Our speech generally comprises mostly of our native language with some filler words coming from another language, which is mostly English. Vice versa of the previous statement is also true as sometimes we speak in English with a few filler words creeping in from our Native language. These mistakes of mixing of languages while speaking is what makes us human and the speech human like.

In general a TTS system consists of 5 stages as shown in Figure 1. These are as follows:

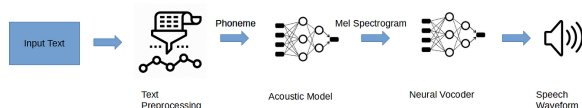


Figure 1: Flow diagram of a TTS System

- **Input Text:** The input from the user in string format.
- **Text Preprocessing:** The input text is passed through a Text Normalizer to convert inputs like currency symbols,

acronyms, etc. into their spoken forms. This spoken form is then passed through the G2P which converts the text into phonemes and then the phonemes to their mapped integers which is the form in which the model can understand.

- **Acoustic Model:** The Acoustic Model takes the phoneme integer numpy array and produces Mel Spectrograms which is a tensor containing elements of float datatype.
- **Neural Vocoder:** The vocoder accepts the output of the acoustic model and produces the waveform tensor containing float type elements.
- **Speech Waveform:** The output from the vocoder is converted to a numpy array which is then converted to .wav file.

In this paper, we try to present a way of achieving the same human like "Code Mix" audio from our Multi-Lingual StyleTTS2[1] which is Style Diffusion based TTS Model, by making changes to the Text Pre-processing layer.

2. Experiment

In our approach we first train a Multi-Lingual StyleTTS2 Model with English(en-US)[2], Spanish(es-ES)[3], Portuguese(pt-PT)[3] and Hindi(hi-IN)¹ Dataset. The total duration of the standardized dataset came up to be approximately 1000 hrs of audio with corresponding 370k utterances. All the datasets listed, were standardized to adhere to the following nuances:

- **Standardized Sample Rate:** Audios resampled to 24 kHz.
- **Standardized Pauses:** Multiple audios joined with 800ms pauses to enhance pauses during punctuations.
- **Silence Padding:** Padding of 400ms on either end of audio to prevent inference noise.
- **Audio Buffering:** All samples less than 1 second in length have been padded at the beginning(on the left side) to increase their duration to greater than 1 second(model limitation) thus, making them viable for training.
- **Dataset Expansion:** All samples having token length more than 512(model limitation) for the transcriptions corresponding to the audios in the dataset have been filtered out.

The first stage of training was run for 50 epochs with the following configurations: max_len as 200, batch_size as 16, sample_rate as 24k, n_token(number of neurons in input layer of the NN) as 184, style_dim as 128, decoder as iSTFTNet[4], lr(learning_rate) as 0.0001.

While for the second stage, which ran for 20 epochs, most of the training hyper-parameters listed above, remained the same with a few minor adjustments and some additional hyper-parameters, which were: batch_size as 2, diff_epoch(style diffusion starting epoch) as 3 and joint_epoch(joint training starting

¹Dataset produced in House

epoch) as 6.

The training was run on A100 GPUs and it ran for about 3.5 weeks, with the first stage taking about half of the time using 8 GPUs while the second stage taking up the rest of the time using 1 GPU(model limitation).

Now, that we have our Multi-Lingual Model trained we went to the inference code, and further to the G2P layer which uses Espeak-ng² that phonemizes the text, before the tokenizer tokenizes them into the corresponding mapped integers called tokens(which is then passed to the TTS Model for inference). This is where we added another layer. We came up with an idea to write a script that recognizes the languages(maximum of 2 for now) present in the input text and then break the text into separate chunks of the corresponding languages. The chunks would then be passed to the corresponding languages' espeak-ng phonemizer which would produce the phonemes. The chunks would then be joined back to make one single phonemized text which would then be passed to the tokenizer to get the tokens that would be passed to the model for inference. A flow diagram with an example text comprising of English and Hindi is shown in Figure 2.

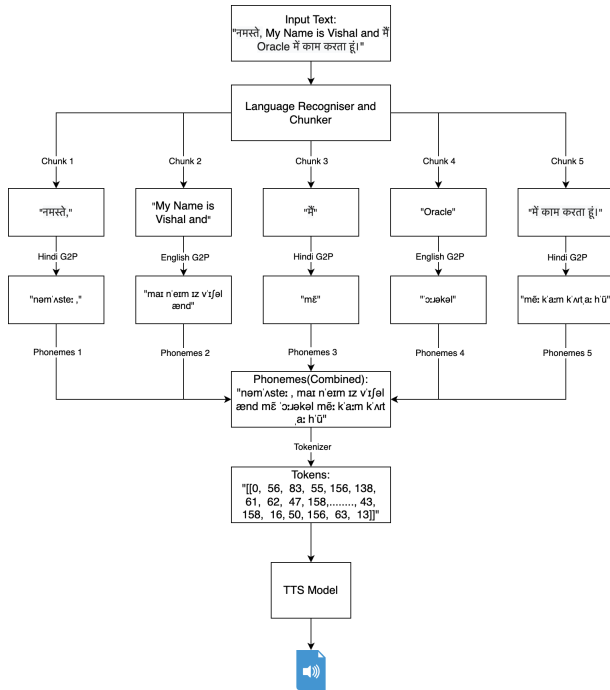


Figure 2: Flow diagram of a Code-Mix TTS System

3. Evaluation

Now, we use this Multi-Lingual StyleTTS2 Model to create an inference pipeline which takes input as text and returns an audio in wav format. Each input text contains of a string in at-most 2 languages, one of them being English and the other being any one out Spanish, Portuguese or Hindi. We then overlay a pipeline for evaluation on top of the inference pipeline and pass 250 said input texts of average length of 210 characters from a custom curated Code-Mix dataset. We run this pipeline on CPU inference mode and evaluate on Automated-MOS(Automated Mean Opinion Score), which is widely regarded as the best comparison standard for TTS models, using Nisqa[5]. On comparison with other industry standard Models, we find that our

²Espeak-ng: <https://github.com/espeak-ng/espeak-ng>

model is performing on par with the them as can be seen in Table 1. The Automated-MOS scores is very much comparable to the StyleTTS2 model trained on a single language(English) dataset, LJSpeech[6] while, like StyleTTS2, it outperforms all other other models like VITS[7] and YourTTS[8].

Table 1: Evaluation of our Code-Mix StyleTTS2 Model

	StyleTTS2 Based TTS Models		Other Models	
Metrics	Our Model	StyleTTS2	VITS	YourTTS
A-MOS(out of 5)	4.12 ± 0.09	4.12 ± 0.08	3.69 ± 0.12	2.35 ± 0.07
Avg Inf Time(secs)	2.88 ± 0.43	2.93 ± 0.22	2.81 ± 0.11	3.25 ± 0.17

4. Conclusion

In this paper, we showcase an approach to alter the G2P layer of any typical TTS model to enhance its capability to process input text in more than one language and generate corresponding, high quality audio output. On subjective(human) evaluation, the audios generated, sound much more natural as compared to single language output as the input text is much more aligned how humans actually speak. There can be a wide range of applications for such a TTS other the simple inference, like creating large training dataset for ASR systems which can help them recognize audios having multi-lingual speech. Future work, on this would be towards recognition and phonemization of text which is mix of more than two languages without the compulsion of one of them being English, as in the case of this paper. Another idea would be to recognize text that are actually spoken in 2 languages but written in a single language like, Hinglish³ or Portunish⁴, etc. The possibilities are endless.

5. References

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³English Text containing Hindi words written in English

⁴Spanish Text containing Portuguese words written in Spanish